

Aggressive Optimization Plan

Goal

Drive higher-impact optimization across runtime-heavy paths and structural service workflows while preserving ledger correctness, reconciliation integrity, and approval/collection behavior.

Scope

The plan is split into two waves:

- **Phase 1:** Runtime performance hotspots
 - **Phase 2:** Structural consolidation and workflow primitives
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Phase 1: Runtime Performance Hotspots

1) Reconciliation batch scans and parity checks

- Refactor reconciliation scan paths to reduce per-row queries and lookup churn.
- Push mismatch filtering to SQL where possible.
- Preserve existing exception codes and entity payload semantics.

Status: Completed

2) Delinquency arrears generation at scale

- Replace per-member/per-period contribution lookups with preloaded contribution maps.
- Reuse a shared unpaid-period builder for single-member and table-generation paths.

Status: Completed

3) Bank-clearing candidate selection and matching cost

- Narrow candidate sets in SQL using amount tolerance and date windows.
- Remove redundant in-memory candidate filtering.

Status: Completed

4) Contribution collection repeated lookups

- Compress repeated household-dependent queries during member-cash increase settlement.
- Reuse preloaded active dependents across household contribution and repayment passes.

Status: Completed

5) Insights aggregate fan-out consolidation

- Replace repeated count/sum/exists fan-out with consolidated aggregate queries.
- Keep dashboard payload shapes and labels stable.

Status: Completed

Phase 2: Structural Consolidation

1) Shared review workflow primitives

- Introduce a common service for operational review metadata updates:

- `status`
- `reviewed_by`
- `reviewed_at`
- `admin_remarks`
- Centralize admin notification fan-out for operational review flows.

Status: Completed

2) Synthetic bank placeholder factory

- Introduce a synthetic statement factory for operational buckets.
- Migrate fund posting and cash-out statement creation callers to shared factory methods.

Status: Completed

3) Centralized clearance linkage payload resolver

- Centralize clear/match linkage payload mapping by flow type:
 - fund posting
 - cash-out
- Add strict unit coverage for resolver output contracts.

Status: Completed

4) Membership approval posting pipeline

- Extract approval posting stages into a dedicated pipeline with explicit stage ordering:
 1. mark approved
 2. prepare cutoff (if CSV-imported)
 3. post subscription fee effects
 4. post opening balances

Status: Completed

Validation Strategy Used

- Run formatter after each substantive pass:
 - `vendor/bin/pint --dirty --format agent`
- Run targeted and dependent regression suites after each phase step, including:
 - bank clearing
 - fund posting
 - cash-out
 - loan delinquency/default
 - membership approval/import-cutoff/subscription-fee flows
 - portal/member insights

Current Plan State

- **Phase 1:** Fully complete
- **Phase 2:** Fully complete